In single-variable calculus, we learned that a differentiable function $y = f(x)$ obtains its local/relative extrema at points in its domain where $f'(x) = 0$ or f' is undefined. However, recall that functions like $g(x) = x^3$ showcase that, just because it has a horizontal tangent line at $x = 0$, does *not* imply the existence of a maxima/minima.

Definition. Let $f(x, y)$ be defined on a region D_f containing the point (a, b). Then $f(a, b)$ is a **local maximum** of f if $f(a, b) \geqslant f(x, y)$ for *all* domain points (x, y) in an open disk centered at (a, b). A **local minimum** is defined analogously.

First Derivative Test for Local Extreme Values. If $f(x, y)$ has a local max/min at an interior point (a, b) in D_f and if the first partial derivatives exist there, then

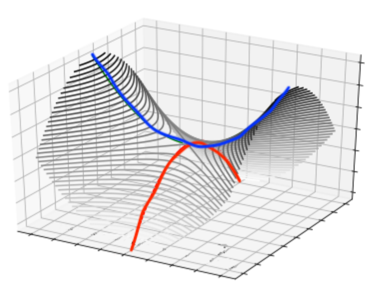
$$f_x(a, b) = 0 \quad \text{and} \quad f_y(a, b) = 0.$$

Definition. Let $f(x, y)$ be defined on the region D_f. Points (a, b) in D_f where

- BOTH f_x and f_y vanish. That is, $\nabla f(a, b) = \langle 0, 0\rangle$;
- AT LEAST ONE of $f_x(a, b)$ or $f_y(a, b)$ do not exist

are called **critical points** of f.

Just like with $x = 0$ on $g(x) = x^3$, just because it is a critical point does NOT mean it is a max/min:



Definition. A differentiable function $f(x, y)$ has a *saddle point* at a critical point (a, b) if every open disk centered at (a, b) has function value both greater than *and* less than $f(a, b)$.

Example. Find any local extrema of $f(x, y) = x^2 + y^2 - 4y + 9$.

Example. Find any local extrema of the function $f(x, y) = y^2 - x^2$.

The saddle point tells us that it is not enough to simply ask for $f_x(a, b) = f_y(a, b) = 0$. Think about $g(x) = x^3$. How do we avoid this function? Well, one way is to ask for functions that maintain their concavity. That is, functions that *curve the same way* all around $f(a, b)$.

Definition. Let $f(x, y)$ be a surface that is differentiable around a point (a, b) in D_f. Suppose further that all of its first and second partial derivatives are continuous around (a, b). Then

$$\mathbf{H}f(x,y) = \mathbf{H}f = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} =$$

is called the **discriminant** or **Hessian** of f.

In practice, the Hessian tells us how 'curvy' the surface is at (a, b):

Second Derivative Test for Local Extreme Values. Suppose $f(x, y)$ and its first and second partial derivatives are continuous around (a, b) in D_f. Suppose further than $f_x(a, b) = f_y(a, b) = 0$.

(i) If $\mathbf{H}f(a, b) > 0$, then the surface curves the same way around $f(a, b)$ (i.e., a local max or min).

- If $f_{xx} < 0$, then it is a local max. (Think: concave down.)
- If $f_{xx} > 0$, then it is a local min. (Think: concave up.)

(ii) If $\mathbf{H}f(a, b) < 0$, then the surface curves up in some directions and down in others around $f(a, b)$ (i.e., a saddle point).

(iii) The test is inconclusive if $\mathbf{H}f(a, b) = 0$.

Example. Find the local extrema of $f(x, y) = 3y^2 - 2y^3 - 3x^2 + 6xy$.